Summary Document

Paper title: An experiment on the effectiveness and efficiency of exploratory testing **Authors:** Wasif Afzal, Ahmad Nauman Ghazi, Juha Itkonen, Richard Torkar, Anneliese Andrews, Khurram Bhatti **Published in:** Empirical Software Engineering, 20(2):844–878, 2015 **DOI:** 10.1007/s10664-014-9301-4

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Aim of the study. Exploratory Testing (ET) is widely practised in industry yet remains under-studied in academia, and the few existing empirical comparisons against documented Test-Case-Based Testing (TCT) are inconclusive. The authors set out to quantify, in a controlled experimental setting, whether testers performing manual functional testing find more (and qualitatively different) defects when using ET than when using TCT, given the same total time budget. They also compare the two approaches with respect to defect-detection difficulty, defect types, defect severity, and the number of false defect reports submitted, in order to provide reliable evidence about the claimed benefits of ET.

Methodology. A controlled experiment was executed in four iterations with 70 subjects in total — 46 high-performing MSc students (top 65% of a Software Verification and Validation course) and 24 industry practitioners from three companies. Subjects were divided into an ET group and a TCT group, balanced across iterations. The System Under Test was jEdit (a Java text editor of ~80 kLOC, used previously by Itkonen et al. 2007), instrumented with 25 seeded faults plus naturally occurring defects. Each subject ran a single 90-minute time-boxed session preceded by a 15-minute briefing; both groups used the same defect-design techniques (equivalence partitioning, BVA, combination testing). The TCT group designed and executed test cases inside the 90 minutes, while the ET group followed a charter and kept session logs. The collected artefacts (defect reports, test logs, design documents) were independently classified for severity (1–3), type, and detection difficulty (4 modes). Hypotheses on defect count, difficulty/type/severity and false-defect counts were tested with Mann–Whitney U at $\alpha=0.05$, and Vargha–Delaney $\hat{A}_{12}$ was used for effect size.

Results. ET found a significantly larger number of defects than TCT in the same time budget, so $H_{0.1}$ was rejected — ET was more *efficient*. ET also detected significantly more defects across every defect-detection-difficulty mode, every defect type (except "documentation"), and every severity level, so $H_{0.2.1}$–$H_{0.2.3}$ were all rejected — ET was more *effective*, particularly at uncovering harder-to-find and more critical defects, while TCT proportionally surfaced more GUI/usability and lower-severity issues. The number of false defect reports was slightly higher for TCT but the difference was *not* statistically significant ($H_{0.3}$ retained). No significant differences were observed between students and practitioners on any of the measures, suggesting the trends hold across populations of comparable domain knowledge.

Implications for research and practice. For research, the study contributes one of the few rigorous controlled experiments on ET, replicating and strengthening Itkonen et al. (2007) by equalising total effort and broadening the subject pool, and it opens follow-ups on feature-coverage measurement, the role of tester experience, ET-vs-automated testing, and the limits of ET when precise regression repeatability is required. For practice, the results push back on the assumption that detailed pre-designed test cases are a prerequisite for effective functional testing: under tight time-boxes, ET delivers more defects of higher severity and difficulty with comparable false-positive rates, making session-based ET a credible primary technique for system-level functional testing — provided testers are skilled and lightweight session logging is in place.